

Program : Diploma in Cyber Forensics and Information security	
Course Code : 4281	Course Title: Programming in Python
Semester : 4	Credits: 4
Course Category: Programme Core Course	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- Provide knowledge in programming using Python language.
- Discover the need of working with strings and functions.
- Illustrate the process of structuring the data using lists, dictionaries, tuples and sets.
- Familiarize exception handling and Object-Oriented Programming concepts.
- Design programs with file operations and database connectivity.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic knowledge about algorithms and programming		Problem Solving and Programming	II
Database concepts		Database Management systems	III

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Utilize the fundamental Python syntax and semantics and control flow statements.	14	Applying
CO2	Construct modular programs using functions and make use of strings handling functions.	15	Understanding Applying

CO3	Demonstrate exception handling and Object-Oriented Programming concepts such as encapsulation, inheritance and polymorphism.	14	Understanding Applying
CO4	Build programs using files and with database connectivity.	15	Understanding Applying
	Series Tests	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	2					
CO3	3	2					
CO4	3	2					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Utilize the fundamental Python syntax and semantics and control flow statements.		
M1.01	Understand basic operators	2	Understanding
M1.02	Explain variables and data types	2	Understanding
M1.03	Understand program flow control blocks	4	Understanding
M1.04	Develop programs using conditional and loop blocks	6	Applying
Contents: Module 1 Introduction to Python: Python variables, Python basic Operators, Understanding python blocks. Python Data Types, Declaring and using Numeric data types. Python Program Flow Control Conditional blocks: if, else and else if, Simple for loops in python, For loop using ranges, string, list and dictionaries. Use of while loops in python, Loop manipulation using pass, continue, break and else. Programming using Python conditional and loop blocks.			

CO2	Construct modular programs using functions and make use of strings handling functions.		
M2.01	Explain Complex data types	2	Understanding
M2.02	Demonstrate String manipulation methods	4	Understanding
M2.03	Demonstrate List manipulation	3	Understanding
M2.04	Build programs using functions.	5	Understanding Applying
	Series Test – I	1	
Contents : Module 2 Python Complex data types: Using string data type and string operations, Defining list and list slicing, Use of Tuple data type. String, List and Dictionary manipulations - string manipulation methods, List manipulation, Dictionary manipulation, Programming using string, list and dictionary, Python Functions - in-built and user defined functions, Organizing python codes using functions.			
CO3	Demonstrate exception handling and Object-Oriented Programming concepts such as encapsulation, inheritance and polymorphism.		
M3.01	Understand Exception handling	1	Understanding
M3.01	Illustrate Exception with Arguments, User-defined Exceptions	2	Understanding
M3.02	Explain Object-Oriented Programming.	2	Understanding
M3.03	Understand Constructor Method	2	Understanding Applying
M3.04	Compare Class Attributes and Data Attributes	2	Understanding Applying
M3.05	Demonstrate Encapsulation, Inheritance and Polymorphism.	5	Understanding Applying
Contents: Module 3 Exceptions-Built-in Exceptions, Handling Exceptions, Exception with Arguments, User-defined Exceptions. Object-Oriented Programming, Classes and Objects, Creating Classes in Python, Creating Objects in Python, Constructor Method, Classes with Multiple Objects, Class Attributes versus Data Attributes, Encapsulation, Inheritance, Polymorphism.			

CO4	Build programs using files and with database connectivity.		
M4.01	Demonstrate Python File Operations	3	Understanding
M4.02	Understand read/write functions	3	Understanding
M4.03	Utilize file pointers	3	Understanding Applying
M4.04	Develop Database Programming	5	Understanding Applying
	Series Test – II	1	
Contents : Module 4 Python File Operations: Reading files, Writing files in python, Understanding read functions, read(), readline(), readlines(). Understanding write functions, write() and writelines() Manipulating file pointer using seek Programming, using file operations.: Connecting to a database, Creating Tables, INSERT, UPDATE, DELETE and READ operations, Transaction Control, Exception Handling in Databases.			

Text / References

T/R	Book Title/Author
T1	Downey, A. et al., "How to think like a Computer Scientist: Learning with Python", John Wiley, 2015
T2	Jeeva Jose & P. Sojan Lal, "Introduction to Computing and Problem Solving with PYTHON", Khanna Publishers, New Delhi, 2016
R1	Wesley J. Chun, "Core Python Applications Programming", 3rd Edition, Pearson Education, 2016
R2	Charles Dierbach, "Introduction to Computer Science using Python", Wiley, 2015

Online Resources

Sl No	Website Link
1	http://coursera.org/course/pythonlearn
2	http://spoken-tutorial.org/